Quantifying comprehensive marine vessel emissions using data fusion

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1 In-line manuscript statistics

1.1 Abstract

- We found that vessels emitted approximately 1.53 billion metric tonnes of CO_2 in 2025. In 2024, the most recent year for which EDGAR data are available, marine vessels accounted for 16% of global fossil fuel emissions from the transportation sector, and 3.6% of global fossil fuel emissions across all sectors.
- Current AIS-only approaches underestimate the magnitude of 2025 emissions by 22%. Meanwhile, because AIS coverage has generally been increasing over time, current AIS-only approaches overestimate the increasing emissions trend by 15% (from 2017 to 2025).

1.2 Introduction

- Climate change mitigation requires accurate accounting across sectors; while the transportation sector as a whole was responsible for 21% of all global CO_2 emissions in 2024, fundamental gaps remain in understanding the contribution by marine vessels.
- For over 770,000 AIS-broadcasting vessels, we quantify emissions for each vessel using precise latitude/longitude/timestamp information for over 119 billion individual AIS messages.
- To account for the vast missing emissions from non-broadcasting vessels (also sometimes known as 'dark vessels'), we develop a machine learning framework that leverages 31.9 million vessel detections from across more than 924,000 S1 scenes to estimate non-broadcasting emissions at a monthly global 1x1-degree gridded resolution, from 2017 through 2025, and disaggregated by vessel type (fishing or non-fishing).

1.3 Results

- Using this data fusion approach, we find that total global CO_2 emissions at sea in 2025 was 1.53 billion MT, with an increase of 38% from 2017 to 2025. If only considering emissions from vessels that broadcast AIS, in line with currently accepted state-of-the-art models, global CO_2 emissions in 2025 would be only 1.19 billion MT, with a more pronounced increase from 2017 to 2025 of 44%.
- Thus using only AIS data from broadcasting vessels underestimates the magnitude of 2025 global CO_2 emissions by 22%, and overestimates the increasing trend for 2017–2025 by 15%
- In 2025, 36% of CO_2 emissions from fishing vessels were from non-broadcasting vessels, and 21% of CO_2 emissions from non-fishing vessels were from non-broadcasting vessels. For both types of vessels, this fraction of emissions from non-broadcasting vessels has generally been declining since 2021, although it increased slightly for fishing vessels in 2025.
- Across the entire 2017 to 2025 time series, almost all non-broadcasting emissions (96.4%) occurred inside the S1 footprint and in pixel-months that were imaged
- Our machine learning model for predicting these emissions directly from observed S1 detection density achieves a high out-of-sample predictive performance (traditional R2 = 0.94 and total percent error of -8.6% for CO_2)
- Only 3.5% of non-broadcasting emissions occurred inside the S1 footprint but in pixel-months that weren't imaged by S1, requiring us to use a two-stage hurdle framework using both a classification model (F1 Score = 0.72) and a regression model (traditional

R2 = 0.73) to predict S1 detection density before predicting emissions (traditional R2 = 0.94).

- Only 0.079% of non-broadcasting emissions occurred outside the S1 footprint entirely, since most vessels operating in these open ocean areas use AIS.
- China, Japan, and United States had the most CO_2 emissions during within-country domestic trips in 2025.
- China, Malaysia, and United States had the most CO_2 emissions during international trips that started or ended in these countries in 2025 (where the emissions from each trip are partitioned equally to its departure and arrival country).
- China, United States, and Indonesia had the most CO_2 emissions during port stays.
- Of the roughly 775,000 AIS-broadcasting vessels in our analysis, only 93,400 (12%) are listed with the IMO, while 45,700 (6%) are listed on some other public registry; this leaves over 636,000 vessels (82%) that have no public registry information and must rely completely on modeled vessel characteristics (main engine power and design speed).
- Because many large vessels are already listed with the IMO, the majority of our total CO_2 emissions come from these vessels (39%); 19% come from vessels listed on other registries; and 42% come from vessels with no public registry information and which rely entirely on modeled vessel characteristics.
- Although vessel activity in the Southern Ocean represents a very small fractions of the world's marine vessel emissions (0.04%), it represents the part of the world experi- encing the highest growth rate of marine vessel emissions (267%).
- These polar estimates are primarily driven by AIS-broadcasting vessels, with only a small fraction coming from non-broadcasting vessels (1.02% in the Southern Ocean).
- Polar regions have extremely limited S1 coverage (only 0.06% of pixel-months in the Southern Ocean were imaged by S1), meaning that trends there may be more susceptible to AIS adoption biases.
- Using the latest data available from the EU's Emissions Database for Global Atmospheric Research (EDGAR), we estimate that in 2024 marine vessels accounted for 16% of all transportation-related CO_2 emissions. Marine vessel CO_2 emissions increased by 32% between 2017 and 2024, 10 times faster than the growth rate of other transportation emissions, and 5 faster than the growth rate of total global emissions across all sectors.

1.4 Discussion

- For non-fishing vessels, the total social cost of CO_2, CH_4, and N_2O emissions in 2021 was 344 billion USD. This represents about 5% of the total value of all goods transported by maritime trade (using 6.63 trillion USD as the 2021 value of maritime-traded goods).

- For fishing vessels, the total social cost of CO_2, CH_4, and N_2O emissions in 2022 was 24 billion USD. This represents about 15% of the total value of all wild capture fisheries (using 159 billion USD as the 2022 value of wild capture fisheries).

1.5 Methods

- The time between pings is typically small (the mean value is 0.0881 hours and the median value is 0.0447 hours, or about 5 minutes and 3 minutes respectively), while the maximum value for this across our dataset is 47 hours (meaning this is the most amount of time over which we will estimate emissions for a single message).
- In all, we analyze AIS data for over 770,000 AIS-broadcasting vessels, across 119 billion individual timestamped AIS messages which are assigned to 14.4 million international trips, 112 million domestic trips, and 96.4 million port visits that occurred between 2017 and 2025.
- As a result, we used a dataset containing 31.9 million vessel detections from 2017 to 2025, of which 14.9 million detections were not matched to a broadcasting vessel ('unmatched'). These detections come from across more than 924,000 individual S1 scenes.
- We were able to use this direct estimation approach for 96.4% of the total estimated non-broadcasting CO_2 emissions .
- Of the total predicted non-broadcasting CO_2 emissions, 3.5% corresponded to indirect estimations within the S1 footprint, while only 0.079% occurred entirely outside the footprint
- For a validation sample of 16,205 vessels, updating vessel metadata with our random forest models substantially improves main engine model performance: traditional R^2 rises from 0.73 using main engine power and design speed from the IMO lookup table, to 0.85 using our random forest predictions, approaching the performance ceiling of 0.9 achieved with the metadata recorded in the registered dataset.
- Validating against the EU MRV database, performance peaks at a traditional R^2 of 0.84 for vessels whose times at sea in our dataset are within 15% of the MRV-reported values (validation sample of 27,657 vessel-year observations). Beyond this threshold, performance declines as discrepancies grow between our annual aggregated trip activity and that recorded in the MRV dataset.

2 Results

2.1 Figure 1

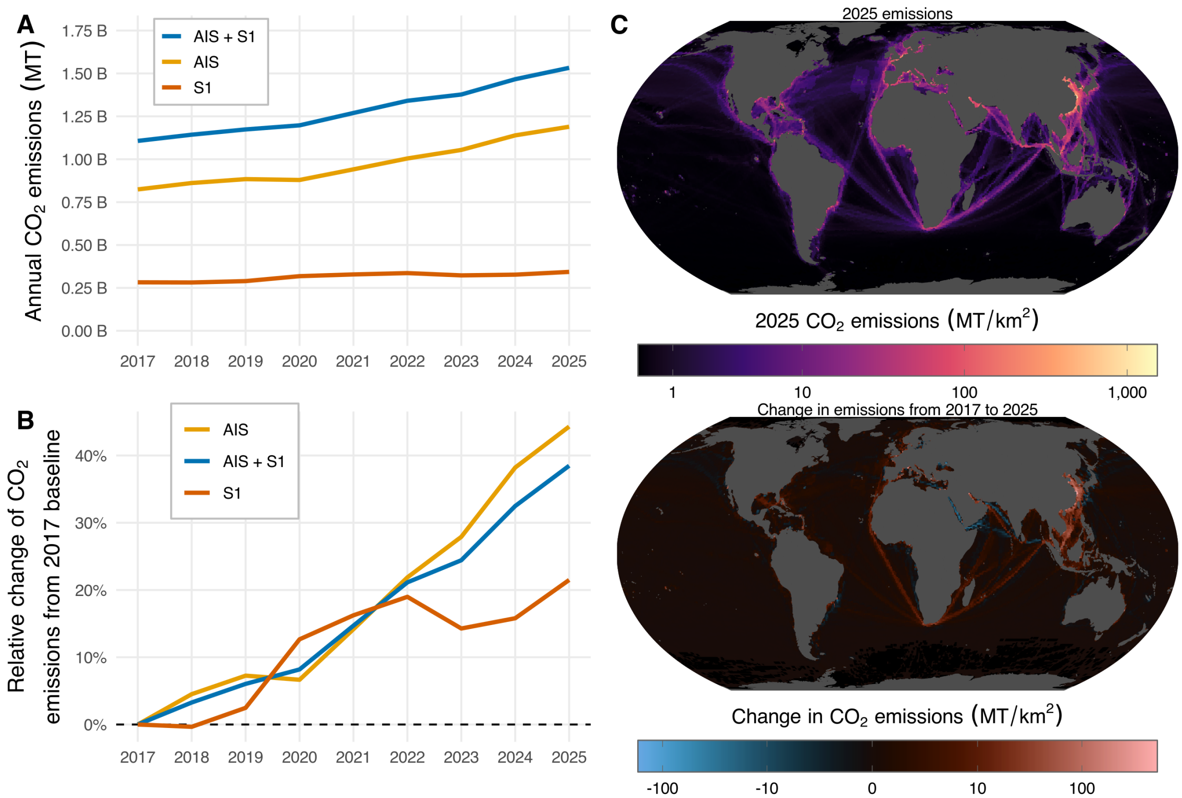


Figure 1: (A) Annual marine vessel CO_2 emissions, with each line colored based on the data source. (B) Relative change in annual marine vessel CO_2 emissions from a 2017 baseline, with each line colored based on the data source. (C) Spatial map of CO_2 emissions in 2025 (top) and change in CO_2 emissions between 2017 and 2025 (bottom), aggregated across AIS-broadcasting vessels and non-broadcasting vessels, and shown at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale).

2.2 Figure 2

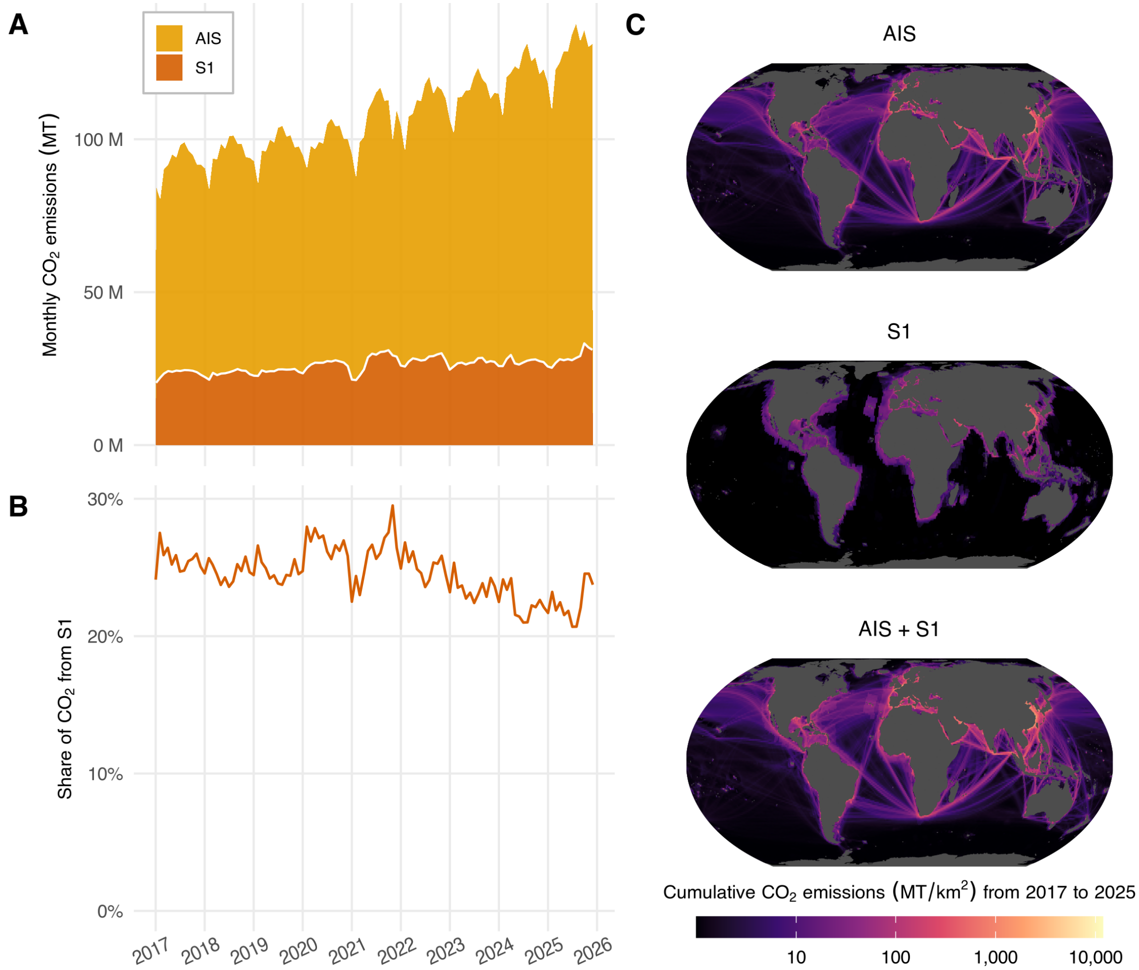


Figure 2: (A) Time series of monthly global CO_2 emissions, shown for both AIS-broadcasting and S1-detected non-broadcasting vessels. (B) Time series of the monthly share of global CO_2 emissions attributable to S1-detected non-broadcasting vessels. (C) Spatial distribution of cumulative 2017-2025 CO_2 emissions from AIS-broadcasting vessels (top), S1-detected non-broadcasting vessels (middle), and both fleets combined (bottom), shown at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale).

2.3 Figure 3

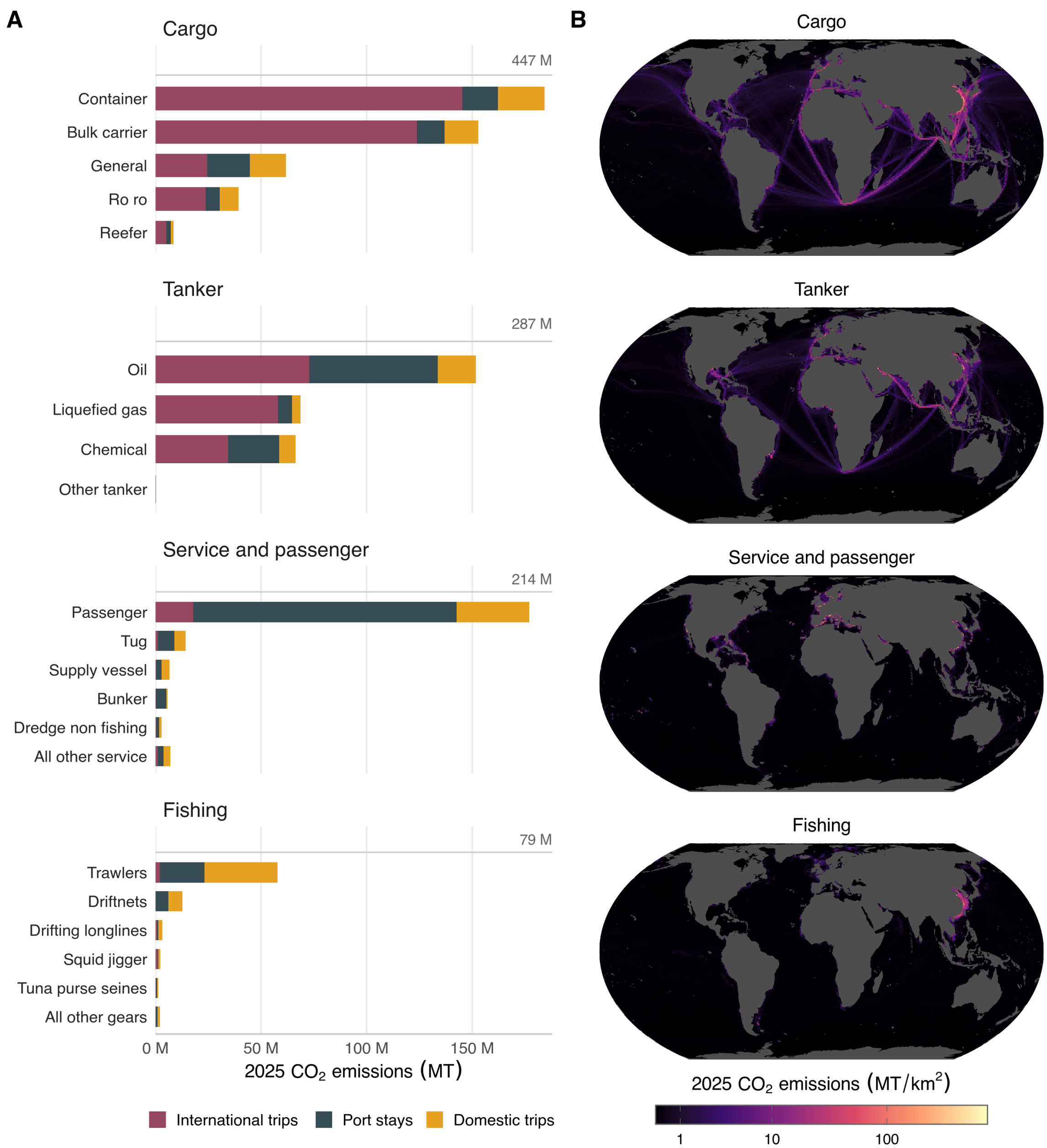


Figure 3: Richness of AIS-broadcasting vessel emissions data, showing: (A) CO_2 emissions by vessel class for AIS-broadcasting vessels during different activity types (international trips, port stays, and domestic trips), aggregated for activities ending in 2025. Vessel classes are grouped into four families, with the total for each family shown at the top right of its panel; within a family the 5 highest-emitting classes are shown individually and any remaining classes are pooled into a single row. (B) Spatial distribution of 2025 CO_2 emissions from AIS-broadcasting vessels for each of those same vessel class families, shown at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale). Panels appear in the same order in both halves of the figure, though panel A covers only emissions assignable to trips and port stays while panel B covers all AIS-broadcasting emissions.

2.4 Figure 4

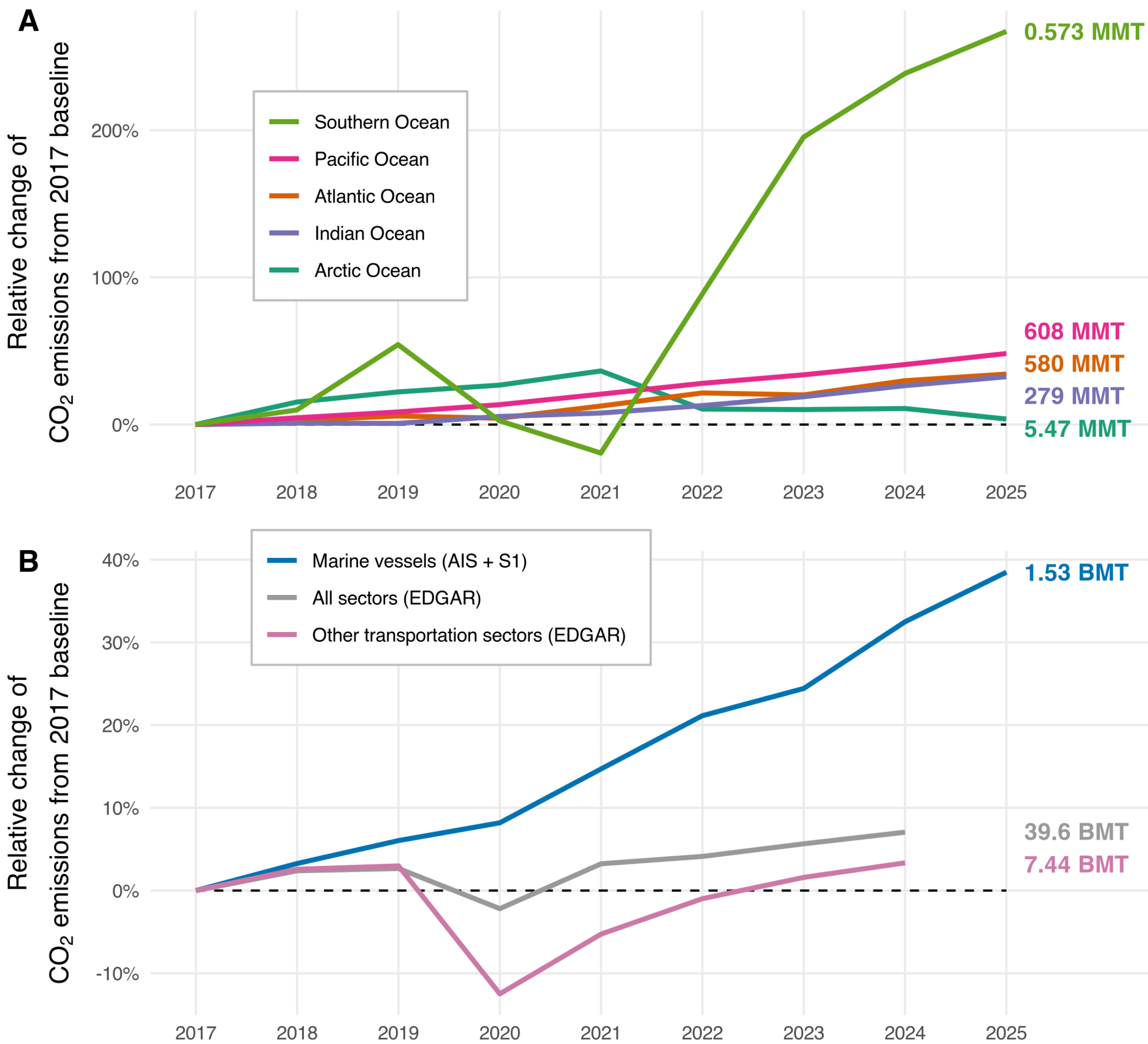


Figure 4: (A) Trend in CO_2 emissions by ocean (normalized to 2017 baseline), using our fused AIS+S1 marine vessel data. (B) Comparison in trend of CO_2 emissions data (normalized to 2017 baseline) between marine transportation emissions from GFW, other transportation emissions from EDGAR, and all sector emissions from EDGAR. On the right-hand side of the figure, absolute values are shown as text and color-coded for each line, for the last year of available data for that series (2025 for the GFW marine vessel series in panel A and B; 2024 for the EDGAR series in panel B).

3 Methods

3.1 Figure 6

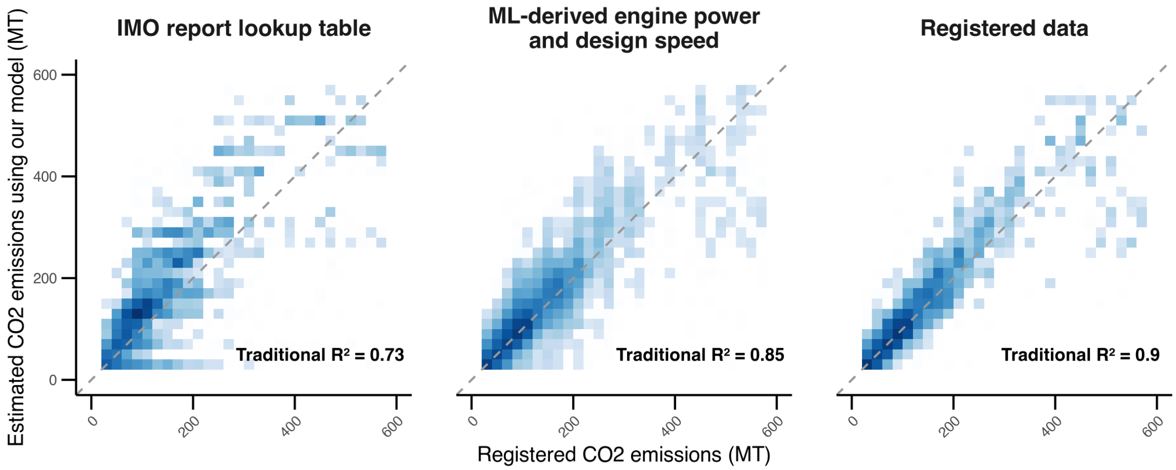


Figure 5: Comparison of daily vessel-level CO_2 emissions estimated using our model with registered emissions from a proprietary vessel database (based on main engine fuel consumption over 24 hours). Estimates are shown using main engine power and design speed from IMO (left), our machine learning random forest models (center), and the same registered dataset (right). The dashed line represents the 1:1 line. Performance values for traditional R^2 are shown for each set of vessel characteristics. Over 16,000 vessels are represented in the figure, and blue shading indicates point density. All data are rule-of-three compliant, meaning each cell (20x20 mt) displayed contains at least three vessels.

Table 1: Performance summary across different thresholds for the hours at sea tolerance between datasets. Results are shown using the R^2 (measuring linear association between observed and predicted values) and traditional R^2 (proportion of variance explained compared to the mean-only baseline).

Tolerance	Number of observations	R^2	Traditional R^2
0.05	11,621	0.84	0.83
0.10	20,432	0.85	0.84
0.15	27,657	0.85	0.84
0.20	33,753	0.85	0.83
0.25	38,980	0.84	0.82
0.30	43,958	0.83	0.81
0.35	48,399	0.83	0.80
0.40	52,490	0.82	0.79

3.2 Table 1

3.3 Figure 7

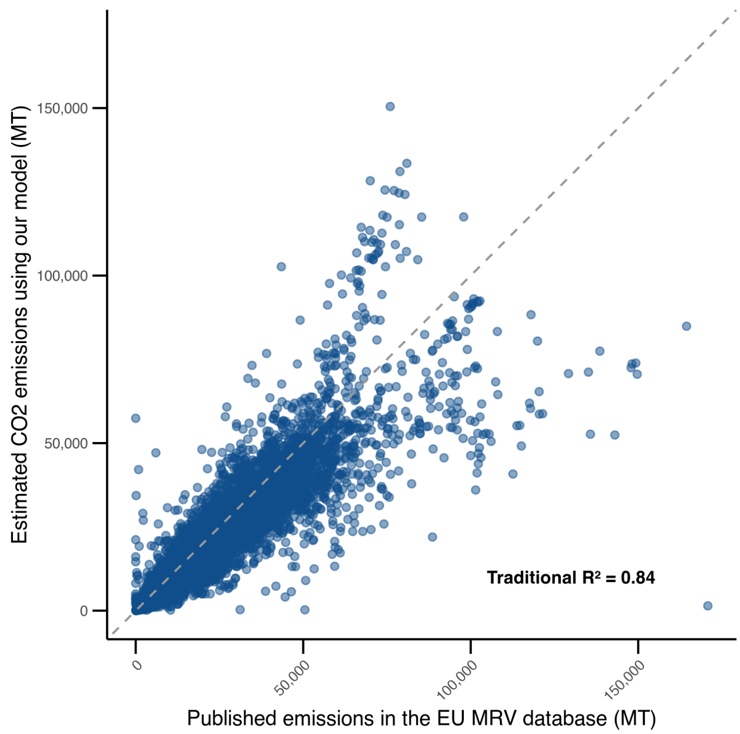


Figure 6: Comparison of annual vessel-level CO_2 emissions estimated using our model with published emissions from the EU MRV database. The sample includes 27,657 observations from yearly aggregated vessel trips where hours at sea differ by no more than 15% between our data and the MRV dataset. The dashed line represents the 1:1 line.

3.4 Figure 8

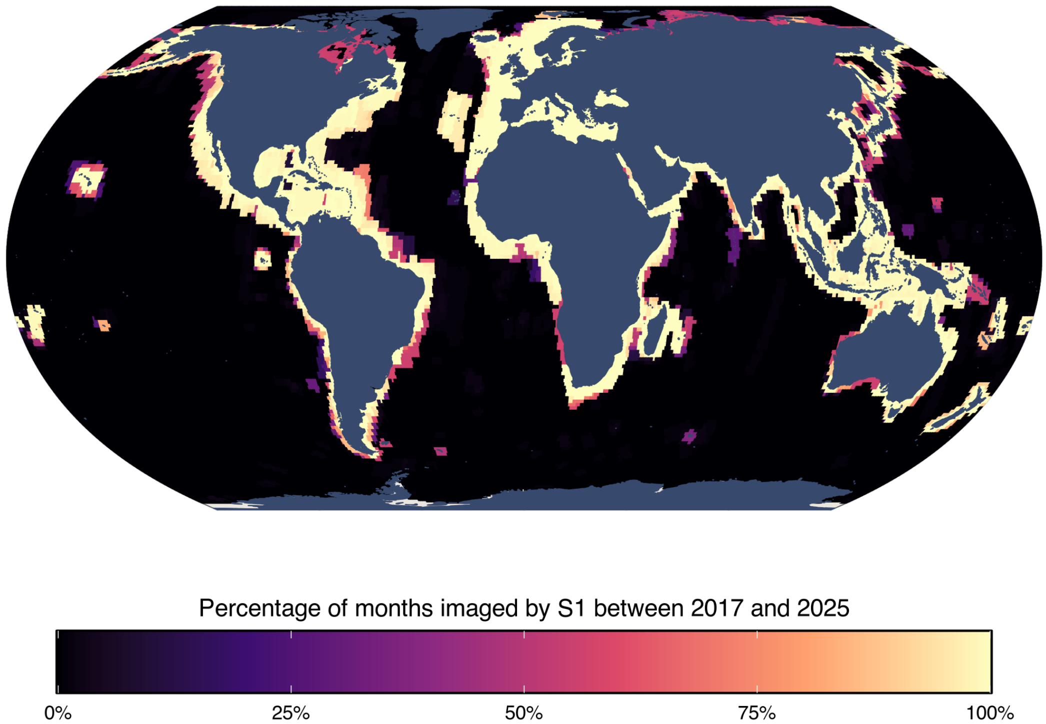


Figure 7: Map showing the percentage of months from 2017 through 2025 that were imaged by S1 for each 1x1 degree pixel.

3.5 Figure 9

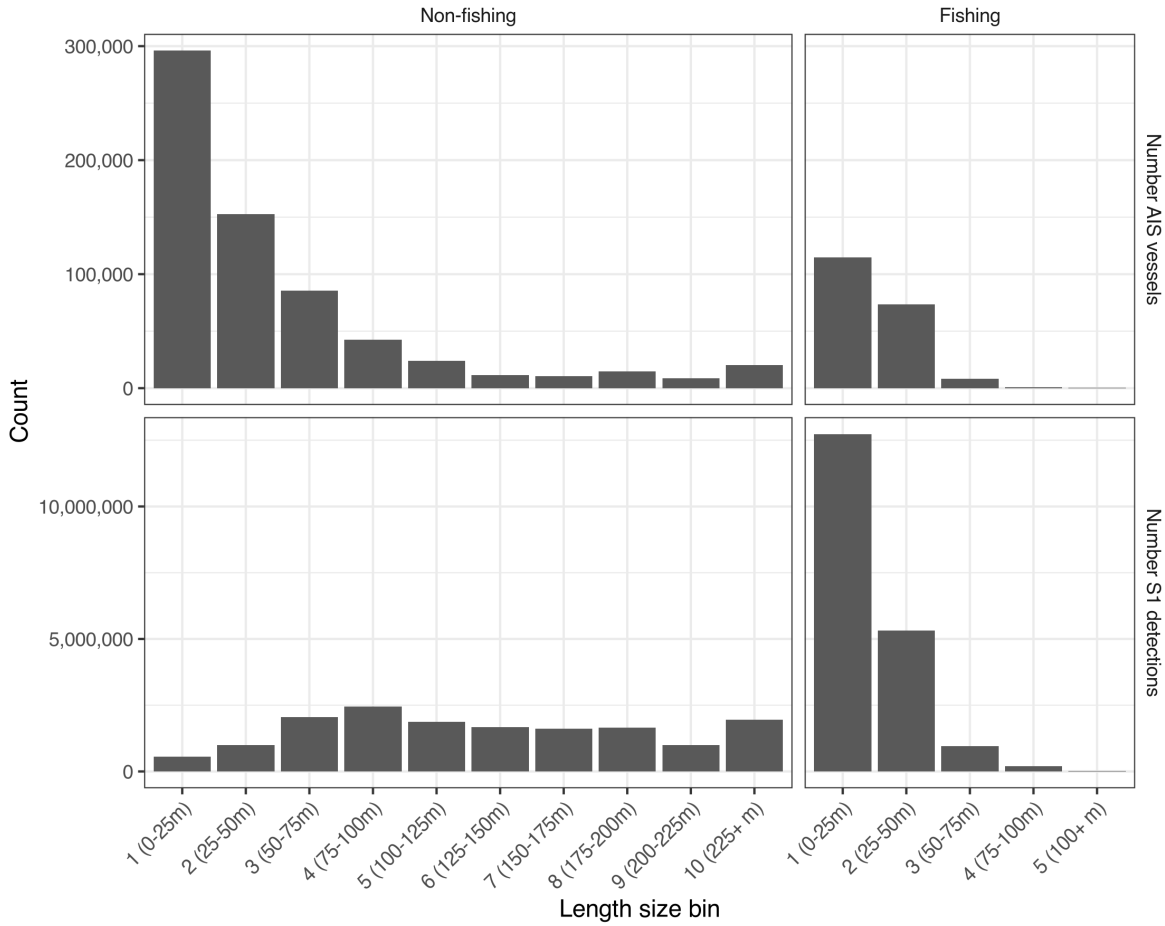


Figure 8: Distributions showing the number of unique AIS vessels and number of S1 detections for each length size bin, disaggregated by fishing and non-fishing vessels.

3.6 Figure 10

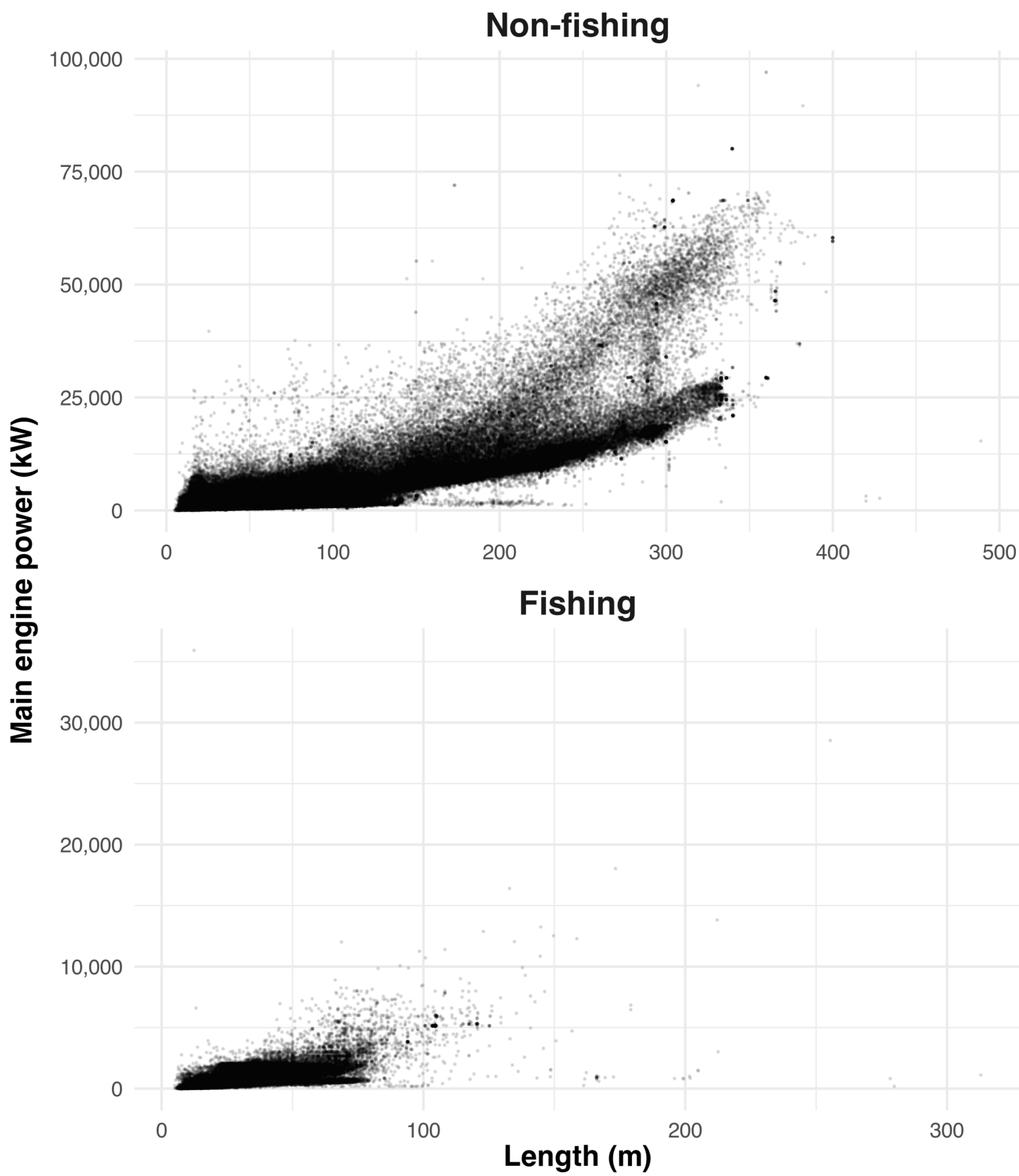


Figure 9: Relationship between main engine power (kilowatts) and length (meters) for all AIS-broadcasting vessels, disaggregated by non-fishing and fishing vessels.

3.7 Table 2

Table 2: Data sources for all included model features, including which random forest model uses the features (M1: detections classification; M2: detections regression; and/or M3: emissions regression), the units, feature type, whether it varies spatially, whether it varies temporally, and the source name and reference citation.

Model feature	Category	M1	M2	M3	Units	Type	Spatial	Temporal	Source	Ref.
Presence/absence of unmatched S1 detections	Dependent variable	Yes	No	No	-	Boolean	Yes	Yes	GFW	[?]
Unmatched S1 detections per km2 imaged	Dependent variable	No	Yes	No	detects/km2	Numeric	Yes	Yes	GFW	[?]
Log(AIS emissions)	Dependent variable	No	No	Yes	log(MT)	Numeric	Yes	Yes	GFW	-
Vessel type (fishing or non-fishing)	Dependent variable	Yes	Yes	Yes	-	Boolean	Yes	Yes	GFW	[?]
Lower threshold of vessel length bin	S1 detection features	Yes	Yes	No	m	Numeric	No	No	GFW	[?]
Vessel length bin (4 categories)	S1 detection features	Yes	Yes	No	-	Factor	Yes	Yes	GFW	[?]
S1 detections per km^2 of area for each length bin	S1 detection features	No	No	Yes	detects/km2	Numeric	Yes	Yes	GFW	[?]
Post-2020 IMO Sulphur policy	Temporal features	No	No	Yes	-	Boolean	No	Yes	IMO	[?]
Ocean basin	Geographic features	Yes	Yes	Yes	-	Factor	Yes	No	GFW	[?]
Pixel water area	Geographic features	Yes	Yes	Yes	km2	Numeric	Yes	No	GFW	[?]
Mean ocean depth	Geographic features	Yes	Yes	Yes	m	Numeric	Yes	No	GFW	[?]
Mean distance from shore	Geographic features	Yes	Yes	Yes	m	Numeric	Yes	No	GFW	[?]
Mean distance from nearest port	Geographic features	Yes	Yes	Yes	m	Numeric	Yes	No	GFW	[?]
Distance to top 20 ports	Geographic features	Yes	Yes	Yes	m	Numeric	Yes	No	Lloyd's List	[?]
Positions reported by AIS type a	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	No	GFW	[?]
Positions reported by AIS type b	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	No	GFW	[?]
Proportion of messages, dynamic AIS receivers	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	Yes	GFW	[?]
Proportion of messages, terrestrial AIS receivers	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	Yes	GFW	[?]
Proportion of messages, satellite AIS receivers	AIS reception quality	Yes	Yes	Yes	-	Numeric	Yes	Yes	GFW	[?]
Mean wind speed	Weather features	No	Yes	No	m/s	Numeric	Yes	Yes	ERA5	[?]
Mean primary wave height	Weather features	No	Yes	No	m	Numeric	Yes	Yes	ERA5	[?]
Mean primary wave period	Weather features	No	Yes	No	s	Numeric	Yes	Yes	ERA5	[?]
Total pixel area imaged by S1 across all time	S1 detection features	Yes	Yes	No	km2	Numeric	Yes	No	GFW	[?]
Pixel water fraction	Geographic features	No	No	Yes	-	Numeric	Yes	No	GFW	[?]
Unique AIS-broadcasting vessels in pixel	AIS reception quality	Yes	Yes	No	-	Numeric	Yes	Yes	GFW	[?]
Unique AIS-broadcasting vessels globally	AIS reception quality	Yes	Yes	No	-	Numeric	No	Yes	GFW	[?]
Month of year	Temporal features	Yes	Yes	Yes	-	Factor	No	Yes	GFW	-
Mean sea surface temperature	Weather features	No	Yes	No	K	Numeric	Yes	Yes	ERA5	[?]
Mean sea level pressure	Weather features	No	Yes	No	Pa	Numeric	Yes	Yes	ERA5	[?]
Total precipitation	Weather features	No	Yes	No	m	Numeric	Yes	Yes	ERA5	[?]

3.8 Figure 11

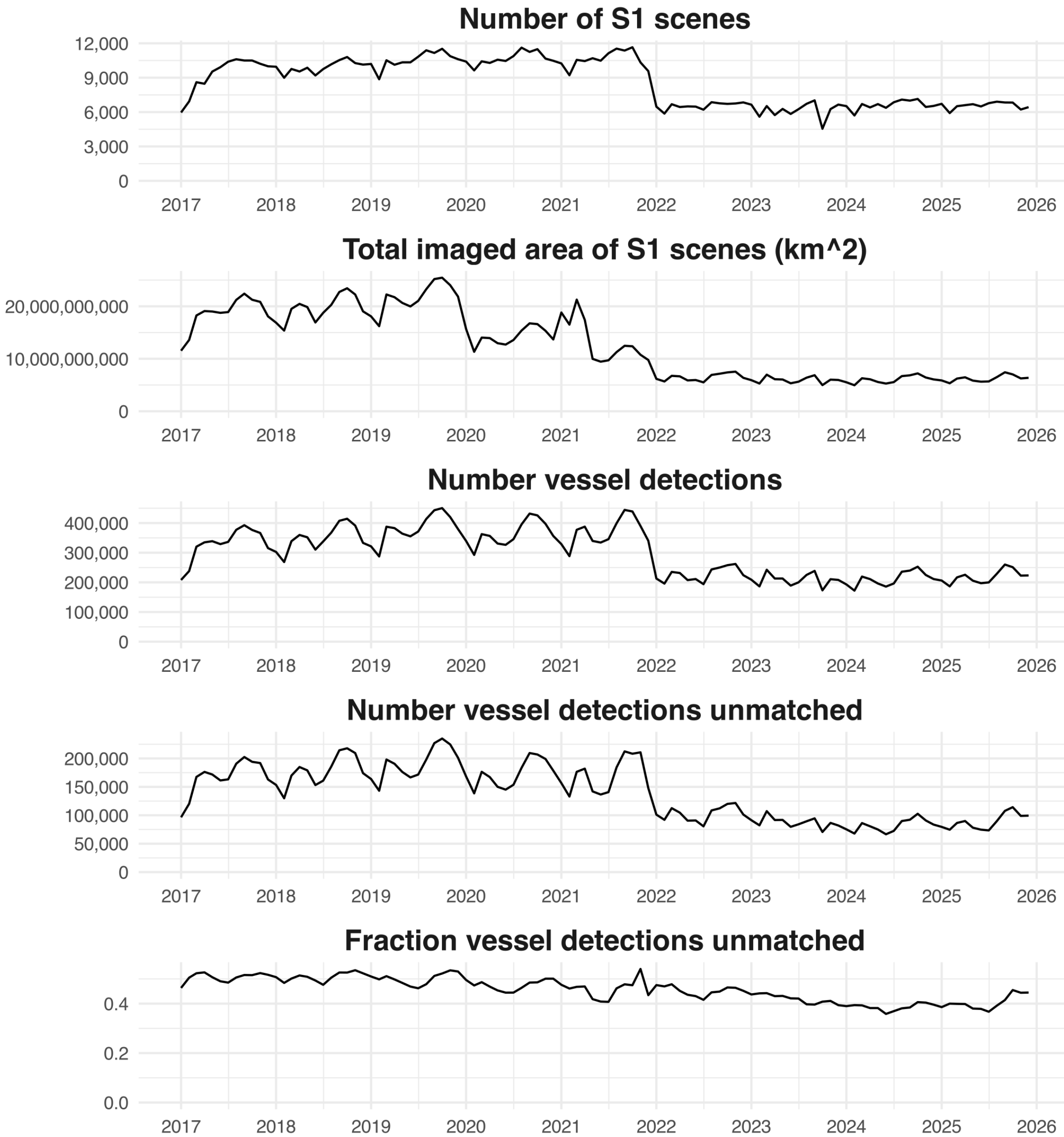


Figure 10: Monthly statistics of global S1 coverage including total number of S1 scenes, total imaged area across all S1 scenes, total number of vessel detections, total number of vessel detections that could not be matched to an AIS-broadcasting vessels, and fraction of the total number of vessel detections that could not be matched to an AIS-broadcasting vessels.

3.9 Figure 12

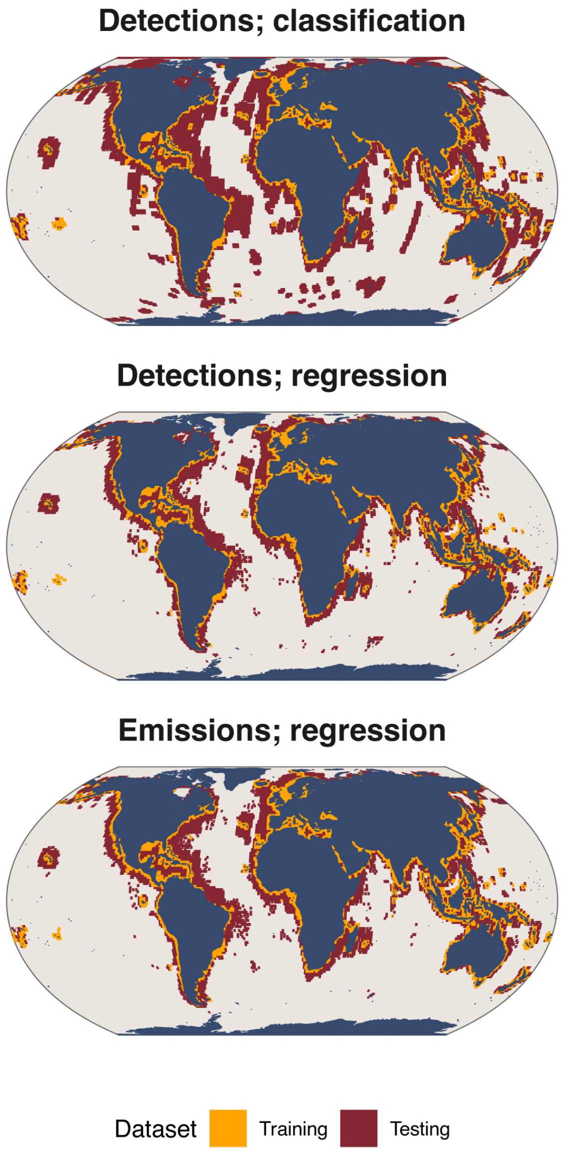


Figure 11: Maps of pixels that were used in the training and testing datasets for the simulated outside S1 footprint performance tests for each model. Pixels <= 100km from the shore were used for training; pixels > 100km from shore were used for testing.

3.10 Figure 13

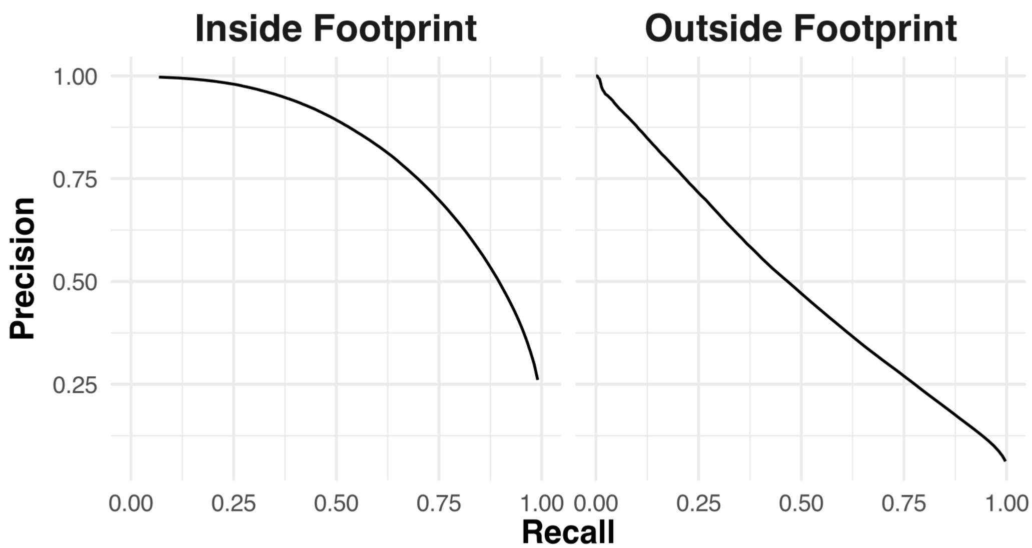


Figure 12: Precision-recall curves of the detection presence/absence classification model, shown for both the inside S1 footprint test and simulated outside S1 footprint test.

3.11 Figure 14

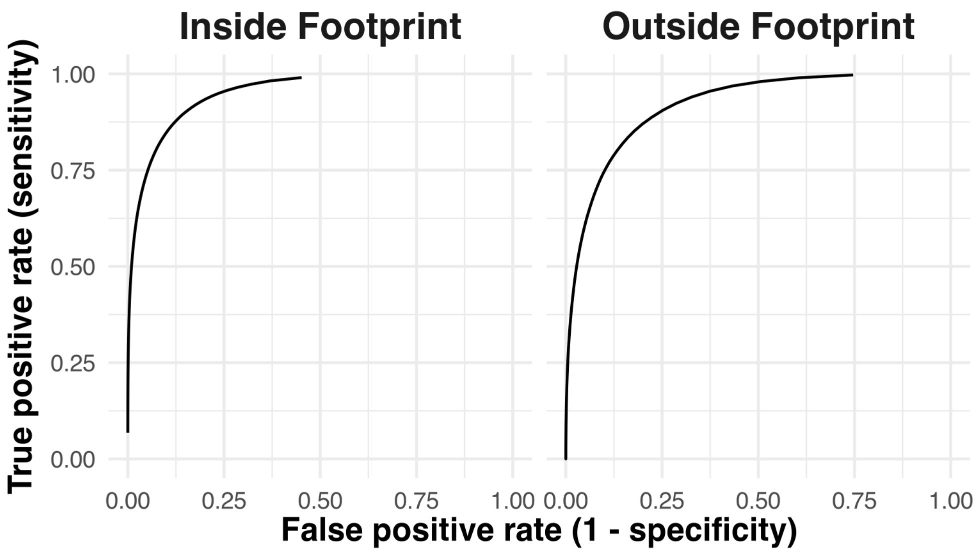


Figure 13: Receiver Operating Characteristic (ROC) curves of the detection classification model, shown for both the inside S1 footprint test and simulated outside S1 footprint test.

3.12 Figure 15

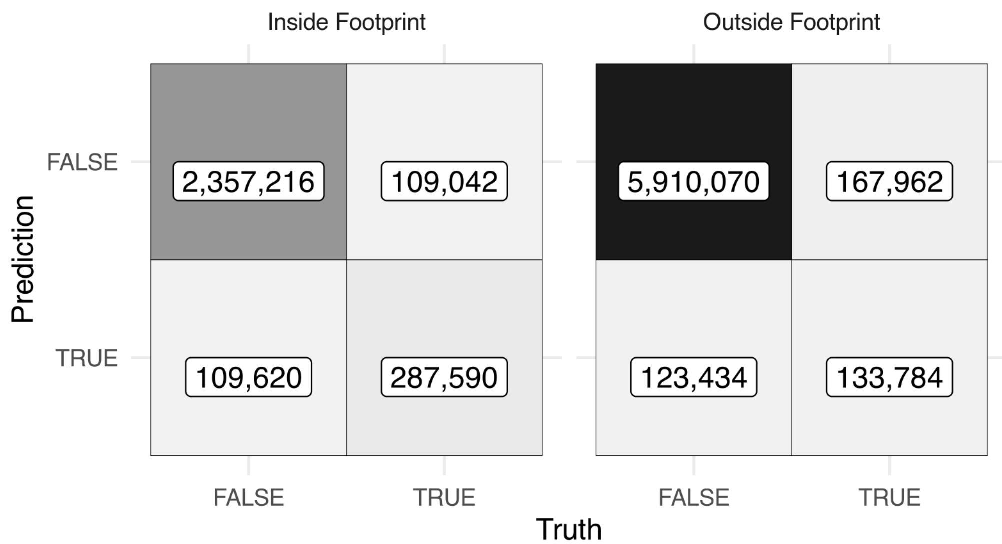


Figure 14: Confusion matrices of the detection classification model, shown for both the inside S1 footprint test and simulated outside S1 footprint test.

Table 3: Summary of dark fleet model performance metrics for the: 1) emissions regression model; 2) detections classification mode; and 3) detections regression model. Performance estimates for each metric are shown for both the inside S1 footprint and outside S1 footprint tests. All metrics are calculated at the unit of observation except for 'Total percent error', which is the percentage error in the total sum of all predictions relative to the total sum of all truth values.

Model	Pollutant	Metric	Inside footprint	Outside footprint
Emissions regression	CO2	Traditional R^2	0.94	0.73
Emissions regression	CH4	Traditional R^2	0.90	0.63
Emissions regression	N2O	Traditional R^2	0.94	0.71
Emissions regression	CO	Traditional R^2	0.92	0.63
Emissions regression	NOX	Traditional R^2	0.87	0.57
Emissions regression	SOX	Traditional R^2	0.95	0.73
Emissions regression	PM2.5	Traditional R^2	0.94	0.65
Emissions regression	PM10	Traditional R^2	0.94	0.65
Emissions regression	VOCs	Traditional R^2	0.88	0.60
Emissions regression	CO2	Total percent error	-8.61	-10.08
Emissions regression	CH4	Total percent error	-20.26	-30.01
Emissions regression	N2O	Total percent error	-10.70	-15.51
Emissions regression	CO	Total percent error	-19.10	-29.29
Emissions regression	NOX	Total percent error	-23.68	-36.44
Emissions regression	SOX	Total percent error	-8.23	-9.01
Emissions regression	PM2.5	Total percent error	-16.21	-25.52
Emissions regression	PM10	Total percent error	-16.21	-25.52
Emissions regression	VOCs	Total percent error	-22.23	-32.45
Detections classification	-	Accuracy	0.92	0.95
Detections classification	-	F1 score	0.72	0.48
Detections classification	-	Precision	0.72	0.52
Detections classification	-	Precision-recall AUC	0.81	0.49
Detections classification	-	Recall	0.73	0.44
Detections classification	-	ROC AUC	0.95	0.92
Detections classification	-	Sensitivity	0.73	0.44
Detections classification	-	Specificity	0.96	0.98
Detections regression	-	Traditional R^2	0.73	0.46
Detections regression	-	Total percent error	1.18	28.46

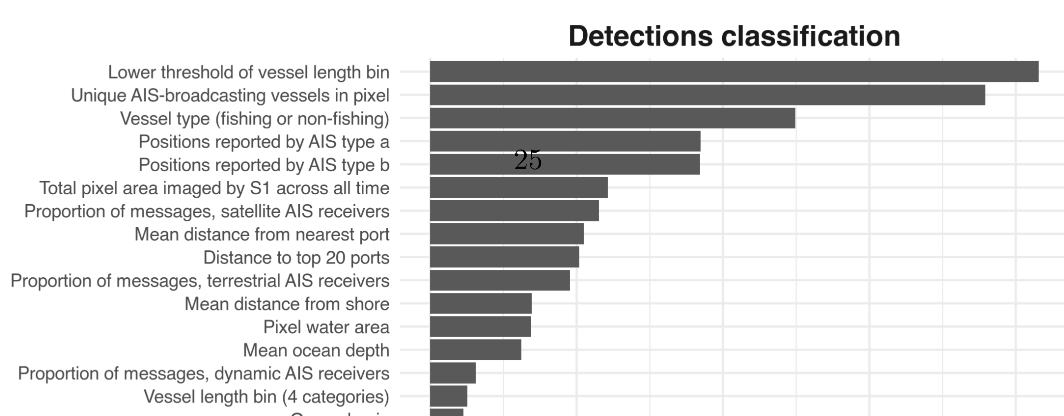
Table 4: Linear regression fit statistics for the final models trained to estimate emissions of non-CO_2 gases from CO_2 emissions. The model for each gas is a zero-intercept linear regression with 5 terms: CO_2 emissions interacted with each combination of period (pre/post-2020 IMO sulphur policy) and vessel type (fishing/non-fishing), plus CO_2 emissions interacted with the log of mean distance from shore.

Pollutant	Term	Estimate	SE	T-stat.	p-value
CH4	CO2 : log(distance from shore + 1)	1.25e-06	3.29e-09	379.9	<0.001
CH4	CO2 : pre-2020 : non-fishing	-1.21e-06	3.17e-08	-38.0	<0.001
CH4	CO2 : post-2020 : non-fishing	-1.92e-06	3.14e-08	-61.3	<0.001
CH4	CO2 : pre-2020 : fishing	3.69e-06	8.42e-08	43.9	<0.001
CH4	CO2 : post-2020 : fishing	3.73e-06	4.52e-08	82.6	<0.001
N2O	CO2 : log(distance from shore + 1)	7.13e-07	2.32e-09	308.0	<0.001
N2O	CO2 : pre-2020 : non-fishing	4.00e-05	2.23e-08	1869.4	<0.001
N2O	CO2 : post-2020 : non-fishing	4.00e-05	2.21e-08	1865.7	<0.001
N2O	CO2 : pre-2020 : fishing	4.00e-05	5.92e-08	727.0	<0.001
N2O	CO2 : post-2020 : fishing	4.00e-05	3.18e-08	1379.9	<0.001
CO	CO2 : log(distance from shore + 1)	6.00e-05	1.57e-07	382.3	<0.001
CO	CO2 : pre-2020 : non-fishing	3.00e-05	1.51e-06	19.2	<0.001
CO	CO2 : post-2020 : non-fishing	-5.49e-06	1.50e-06	-3.7	<0.001
CO	CO2 : pre-2020 : fishing	2.70e-04	4.02e-06	67.9	<0.001
CO	CO2 : post-2020 : fishing	2.80e-04	2.16e-06	128.1	<0.001
NOX	CO2 : log(distance from shore + 1)	2.02e-03	4.33e-06	466.7	<0.001
NOX	CO2 : pre-2020 : non-fishing	-7.34e-03	4.00e-05	-176.3	<0.001
NOX	CO2 : post-2020 : non-fishing	-8.45e-03	4.00e-05	-205.2	<0.001
NOX	CO2 : pre-2020 : fishing	-9.70e-04	1.10e-04	-8.8	<0.001
NOX	CO2 : post-2020 : fishing	-2.14e-03	6.00e-05	-36.1	<0.001
SOX	CO2 : log(distance from shore + 1)	-1.00e-05	7.15e-08	-173.1	<0.001
SOX	CO2 : pre-2020 : non-fishing	6.30e-03	6.88e-07	9164.2	<0.001
SOX	CO2 : post-2020 : non-fishing	1.66e-03	6.81e-07	2441.1	<0.001
SOX	CO2 : pre-2020 : fishing	6.37e-03	1.83e-06	3486.1	<0.001
SOX	CO2 : post-2020 : fishing	1.67e-03	9.81e-07	1705.6	<0.001
PM2.5	CO2 : log(distance from shore + 1)	4.00e-05	9.11e-08	401.0	<0.001
PM2.5	CO2 : pre-2020 : non-fishing	4.10e-04	8.76e-07	472.5	<0.001
PM2.5	CO2 : post-2020 : non-fishing	5.00e-05	8.67e-07	58.1	<0.001
PM2.5	CO2 : pre-2020 : fishing	5.30e-04	2.33e-06	228.9	<0.001
PM2.5	CO2 : post-2020 : fishing	2.10e-04	1.25e-06	168.4	<0.001
PM10	CO2 : log(distance from shore + 1)	4.00e-05	9.90e-08	401.0	<0.001
PM10	CO2 : pre-2020 : non-fishing	4.50e-04	9.52e-07	472.5	<0.001
PM10	CO2 : post-2020 : non-fishing	5.00e-05	9.43e-07	58.1	<0.001
PM10	CO2 : pre-2020 : fishing	5.80e-04	2.53e-06	228.9	<0.001
PM10	CO2 : post-2020 : fishing	2.30e-04	1.36e-06	168.4	<0.001
VOCs	CO2 : log(distance from shore + 1)	1.00e-04	2.02e-07	495.7	<0.001
VOCs	CO2 : pre-2020 : non-fishing	-3.90e-04	1.94e-06	-203.0	<0.001
VOCs	CO2 : post-2020 : non-fishing	-4.40e-04	1.92e-06	-228.0	<0.001
VOCs	CO2 : pre-2020 : fishing	-2.00e-04	5.15e-06	-39.0	<0.001
VOCs	CO2 : post-2020 : fishing	-2.10e-04	2.77e-06	-75.7	<0.001

3.13 Table 3

3.14 Table 4

3.15 Figure 16



3.16 Figure 17

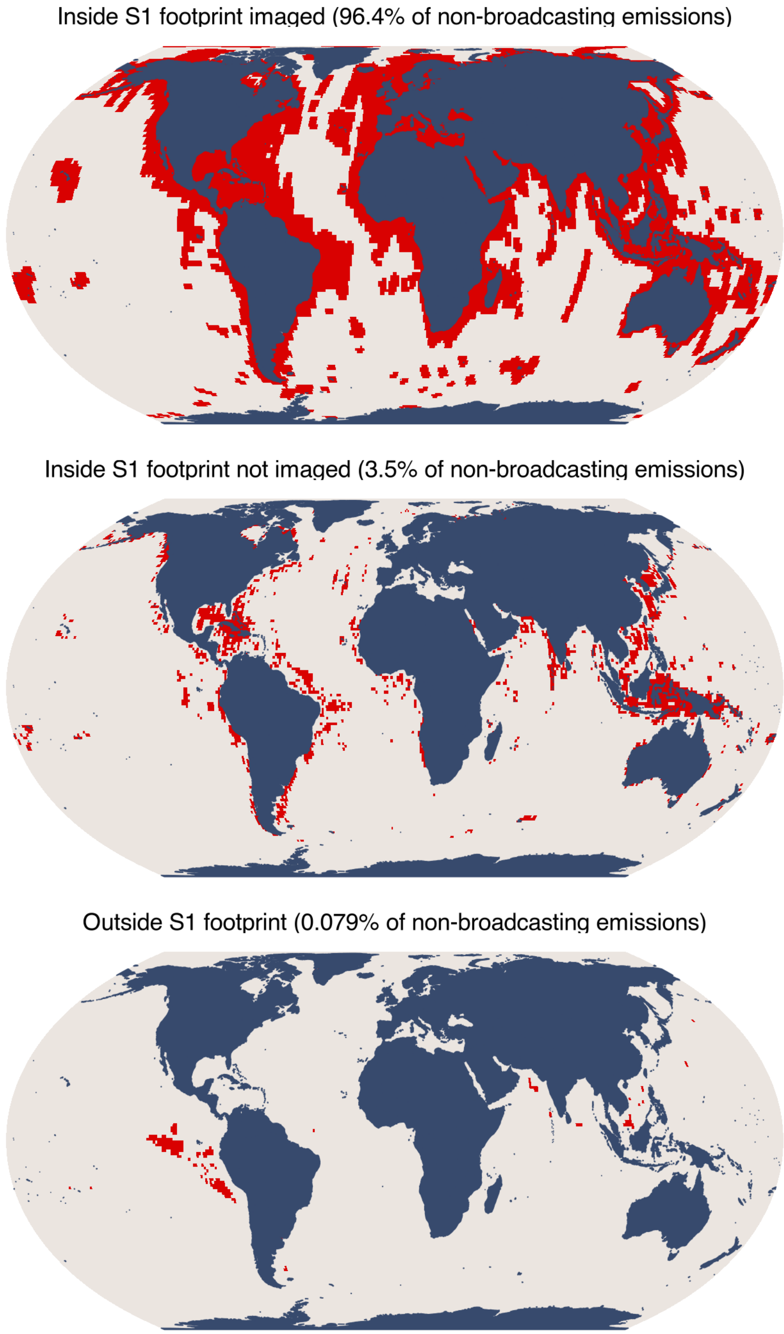


Figure 16: Spatial coverage of the non-broadcasting emissions model. Areas are categorized according to whether they fall inside or outside the S1 footprint and whether they have been imaged by S1 at any point in time. A given pixel may be imaged in some months but not in others. Pixels outside the footprint have never been imaged. The percentage of total CO_2 emissions from non-broadcasting vessels attributed to each category is shown.

4 Supplementary materials: Supplementary figures

4.1 Figure S1

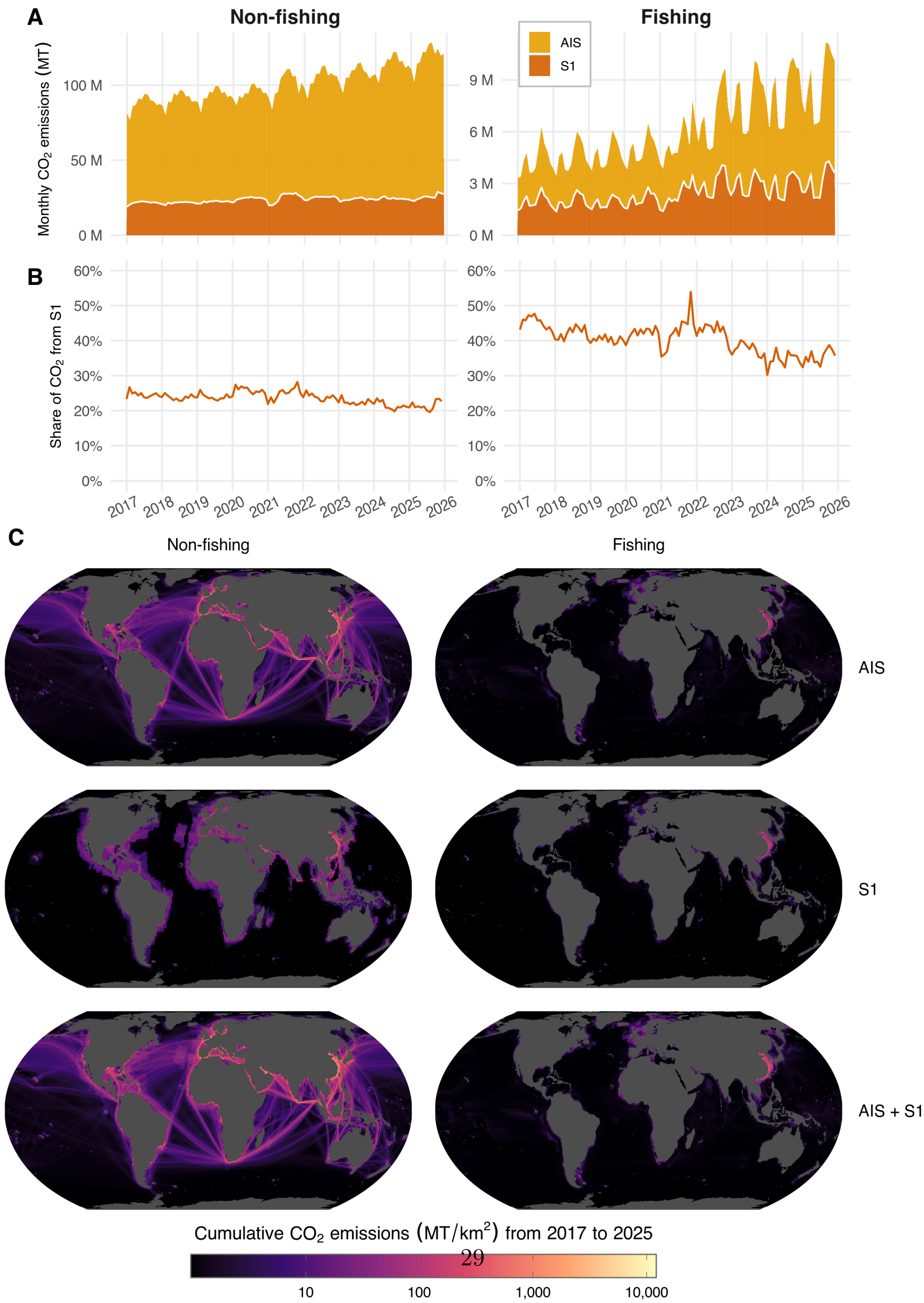


Figure 17: (A) Time series of monthly global CO_2 emissions, shown for both AIS-broadcasting and S1-detected non-broadcasting vessels, and for both fishing and non-fishing vessels. (B) Time series of the monthly share of global CO_2 emissions attributable to S1-detected non-broadcasting vessels, for both fishing and non-fishing vessels. (C) Spatial distribution of cumulative 2017-2025 CO_2 emissions from AIS-broadcasting

4.2 Figure S2

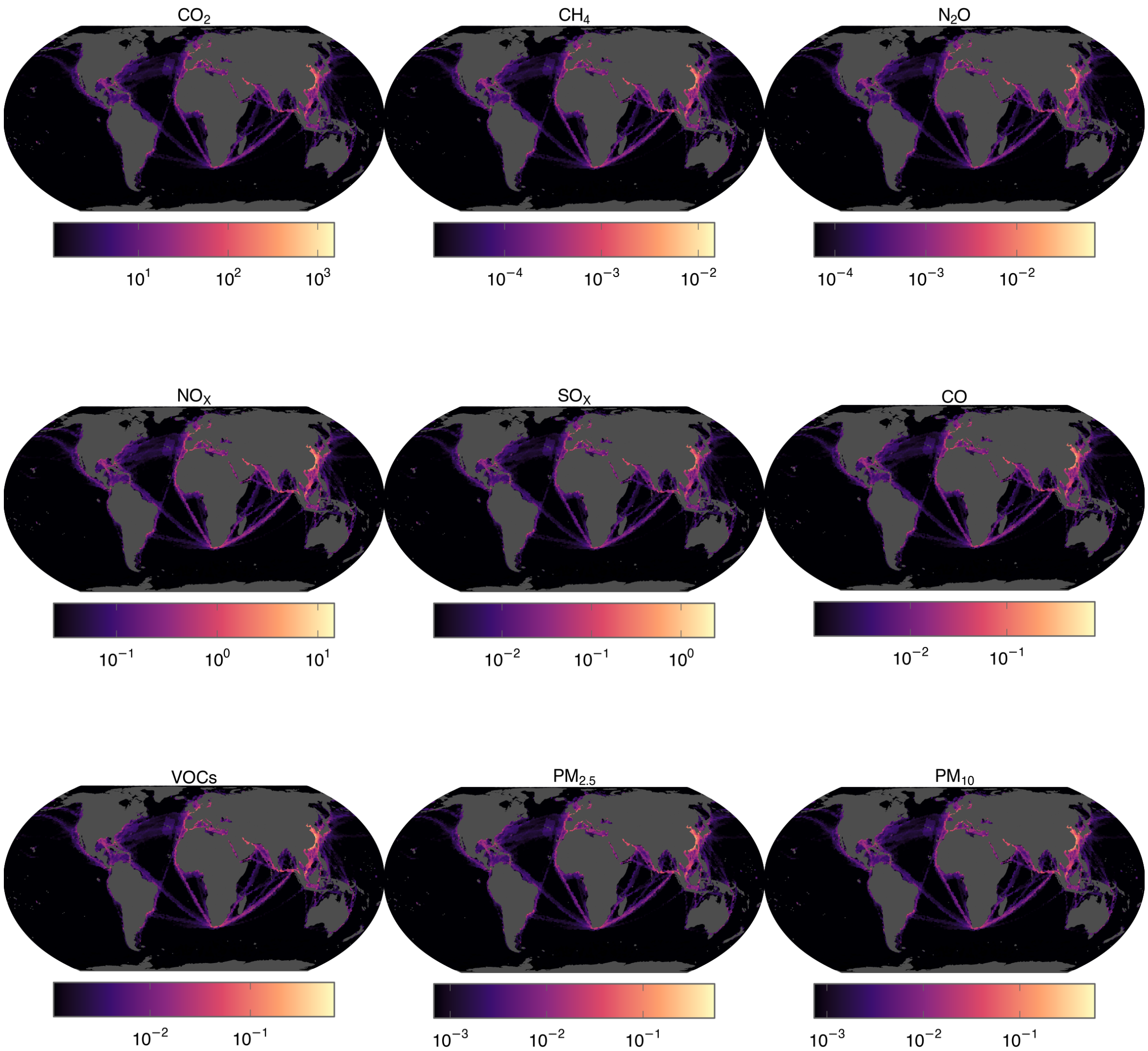


Figure 18: Spatial maps of 2025 emissions for each GHG and pollutant, aggregated across AIS-broadcasting vessels and non-broadcasting vessels, and shown at a 1x1 degree spatial resolution. Emissions are area-normalized for each pixel (metric tonnes per square kilometer) and shown using a log-10 scale. Each panel is scaled independently, spanning the 75th percentile to the maximum of that pollutant, with lower values drawn in the darkest colour; colours are therefore not comparable between panels.

4.3 Figure S3

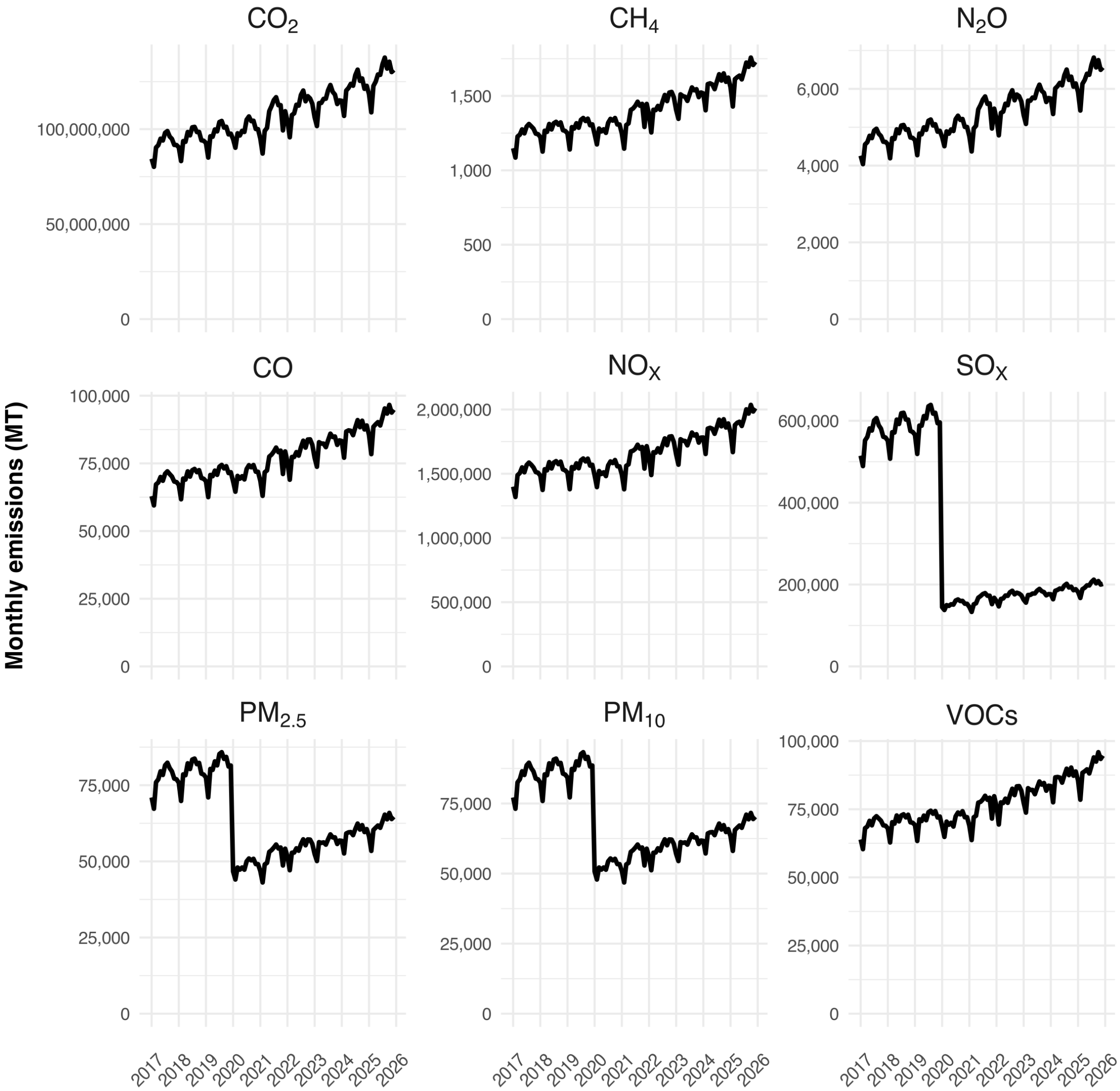


Figure 19: Time series of monthly global emissions for each GHG and pollutant, aggregated across AIS-broadcasting vessels and non-broadcasting vessels for which we only have Sentinel-1 data.

4.4 Figure S4

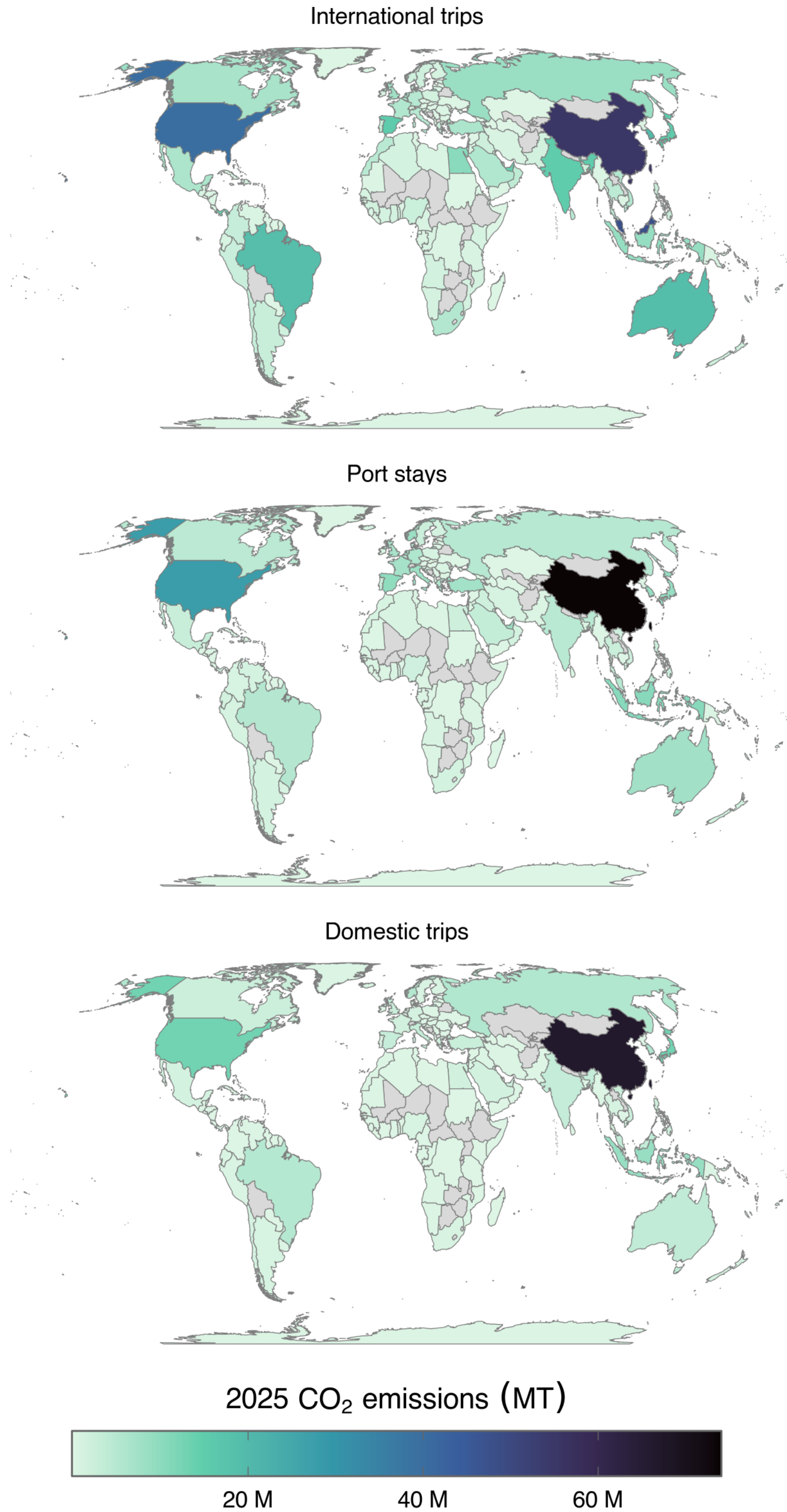


Figure 20: CO_2 emissions by country for AIS-broadcasting vessels during different activity types (international trips, port stays, and domestic trips), aggregated for activities ending in 2025. Emissions for each international trip are partitioned equally to its departure and arrival country. Countries with no values are shown in gray.

4.5 Figure S5

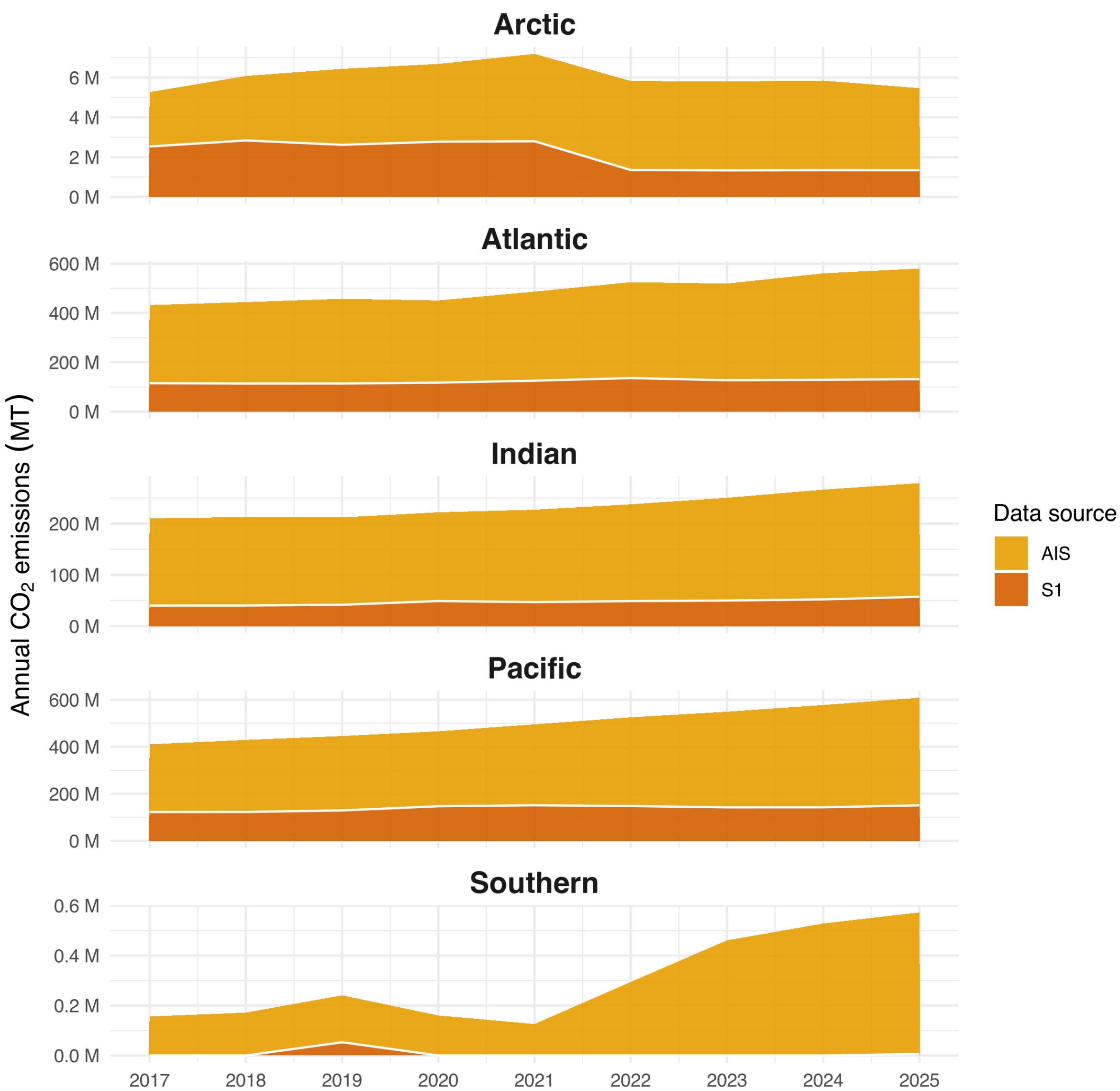


Figure 21: Annual total CO2 emissions by ocean, disaggregated by data source (AIS-broadcasting vessels, and non-broadcasting vessels detected by S1).

5 Supplementary materials: Supplementary tables

5.1 Table S1

Table 5: Summary of 2017 and 2025 emissions, data source and pollutant.

Source	Pollutant	2017 emissions, MT	2025 emissions, MT	Percent 2025 total	2017 to 2025 change
AIS	CO2	8.24e+08	1.19e+09	78%	44%
S1	CO2	2.83e+08	3.43e+08	22%	21%
AIS + S1	CO2	1.11e+09	1.53e+09	100%	38%
AIS	CH4	11,500	15,800	80%	37%
S1	CH4	3,420	3,990	20%	17%
AIS + S1	CH4	14,900	19,700	100%	33%
AIS	N2O	41,800	59,500	78%	42%
S1	N2O	13,900	16,800	22%	21%
AIS + S1	N2O	55,700	76,300	100%	37%
AIS	CO	628,000	865,000	80%	38%
S1	CO	189,000	222,000	20%	17%
AIS + S1	CO	817,000	1,090,000	100%	33%
AIS	NOX	14,100,000	18,500,000	80%	31%
S1	NOX	3,960,000	4,490,000	20%	14%
AIS + S1	NOX	18,100,000	2.3e+07	100%	27%
AIS	SOX	5,030,000	1,830,000	78%	-64%
S1	SOX	1,750,000	527,000	22%	-70%
AIS + S1	SOX	6,770,000	2,360,000	100%	-65%
AIS	PM2.5	701,000	590,000	79%	-16%
S1	PM2.5	226,000	153,000	21%	-32%
AIS + S1	PM2.5	927,000	743,000	100%	-20%
AIS	PM10	762,000	641,000	79%	-16%
S1	PM10	246,000	166,000	21%	-32%
AIS + S1	PM10	1,010,000	807,000	100%	-20%
AIS	VOCs	641,000	868,000	80%	35%
S1	VOCs	185,000	213,000	20%	15%
AIS + S1	VOCs	825,000	1,080,000	100%	31%

5.2 Table S2

Table 6: Underestimation of 2025 emissions using only AIS data, and overestimation of 2017 to 2025 emissions trend using only AIS data, by pollutant.

Pollutant	Underestimation of 2025 emissions	Overestimation of 2017-2025 emissions trend
CO2	22%	15%
CH4	20%	15%
N2O	22%	15%
CO	20%	14%
NOX	20%	14%
SOX	22%	-2%
PM2.5	21%	-20%
PM10	21%	-20%
VOCs	20%	15%

Table 7: For each ocean, summary of: 2025 total (AIS+S1) CO_2 emissions (metric tonnes); percentage of global total (AIS+S1) 2025 CO_2 emissions from across all oceans; percent of 2025 total (AIS+S1) CO_2 emissions by non-broadcasting vessels; percent change in total (AIS+S1) CO_2 emissions from 2017 to 2025; and the percentage of all observations in the dataset (pixel-months) from 2017-2025 that were imaged by S1.

Summary statistic	Atlantic	Southern	Pacific	Indian	Arctic
2025 total CO2 emissions (million MT)	449	0.567	457	222	4.13
Percent of total CO2 2025 emissions	39.37%	0.04%	41.28%	18.93%	0.37%
Percent of total CO2 emissions by non-broadcasting	22.64%	1.02%	24.81%	20.58%	24.50%
Percent change in total CO2 emissions (2017 to 2025)	34.31%	267.19%	48.27%	32.49%	3.74%
Percent observations imaged by S1 (across 2017-2025)	35.94%	0.06%	17.74%	16.99%	13.55%

5.3 Table S3

5.4 Table S4

Table 8: For each data source, summary of: total 2024 CO_2 emissions (metric tonnes); and percent change in CO_2 emissions from 2017 to 2024.

Data source	2024 emissions, billion MT	Change in emissions, 2017 to 2024
Other transportation sectors (EDGAR)	7.44	3.35%
All sectors (EDGAR)	39.6	7.06%
Marine vessels (AIS + S1)	1.47	32.49%

5.5 Table S5

Table 9: For both non-fishing vessels and fishing vessels, we summarize: the total emissions (metric tonnes) of each GHG gas; the social cost associated with each GHG (USD/metric tonne); the social cost of each (USD); the total social cost summed across all three gases (USD). We use 2021 for non-fishing vessels since it is the year for which we have the total value of all traded goods at sea. We use 2022 for fishing vessels since it is the year for which we have the total value of wild capture fisheries.

Year	Vessel type	GHG	Emissions (MT)	SC (USD/MT)	SC (billion USD)
2021	Non-fishing	CH4	15,296	2,900	0.044
2021	Non-fishing	CO2	1,205,568,569	283	341.176
2021	Non-fishing	N2O	60,152	49,600	2.984
Total	Non-fishing	-	-	-	344.204
2022	Fishing	CH4	1,373	2,900	0.004
2022	Fishing	CO2	82,662,425	283	23.393
2022	Fishing	N2O	4,211	49,600	0.209
Total	Fishing	-	-	-	23.606

6 Supplementary materials: Supplementary figures: Comparing our emission estimates to other inventories

6.1 Figure S6

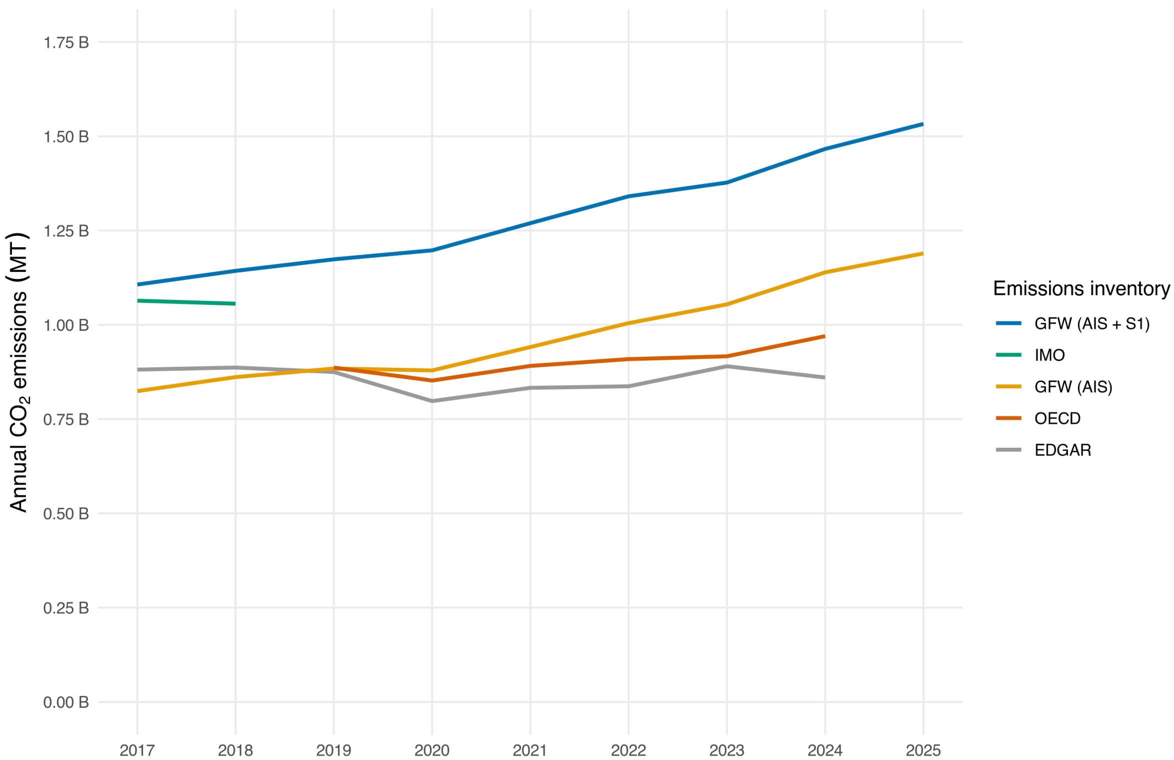


Figure 22: Comparison of annual marine vessel CO_2 emissions inventory estimates.

Table 10: Top 30 countries by 2025 CO_2 emissions (millions of metric tonnes) from AIS-broadcasting vessels, shown separately for each activity type, with all remaining countries pooled into a single Other row. Emissions for each international trip are partitioned equally to its departure and arrival country. These are the totals mapped in Figure ??.

International trips		Port stays		Domestic trips	
Country	CO_2	Country	CO_2	Country	CO_2
China	55.5	China	74.0	China	66.8
Malaysia	48.1	United States	27.8	Japan	13.1
United States	39.9	Indonesia	10.6	United States	13.1
South Korea	20.2	Spain	9.8	Indonesia	8.8
Australia	19.1	South Korea	9.0	South Korea	5.7
Brazil	18.9	Netherlands	9.0	Russia	5.6
Spain	15.7	Japan	8.9	Norway	5.4
Japan	15.5	United Arab Emirates	8.4	Brazil	5.2
India	15.2	Norway	8.4	Australia	3.8
United Arab Emirates	14.2	Italy	8.1	Malaysia	3.4
Panama	13.8	Australia	7.6	India	3.4
Singapore	12.7	Turkey	7.4	Italy	3.2
Egypt	10.4	France	7.2	Spain	3.1
Indonesia	9.0	United Kingdom	6.7	Turkey	2.7
Russia	8.5	Germany	6.0	United Kingdom	2.6
United Kingdom	7.7	Greece	5.8	Canada	2.5
Netherlands	7.4	Brazil	5.3	Vietnam	2.3
Canada	6.9	Russia	5.0	United Arab Emirates	2.2
Mexico	6.4	India	4.8	Philippines	2.2
Turkey	6.3	Malaysia	4.7	Greece	1.8
Taiwan	6.1	Canada	4.5	Taiwan	1.8
Belgium	6.0	Denmark	4.4	France	1.7
France	5.6	Belgium	3.9	Egypt	1.7
Saudi Arabia	5.5	Saudi Arabia	3.3	Saudi Arabia	1.6
South Africa	5.5	Mexico	2.8	Mexico	1.3
Vietnam	5.4	Thailand	2.8	Germany	1.1
Italy	5.0	Sweden	2.5	Chile	1.1
Germany	5.0	Philippines	2.3	Thailand	1.1
Sri Lanka	4.9	Taiwan	2.2	New Zealand	1.0
Morocco	4.7	Egypt	2.2	Argentina	1.0
Other	107.2	Other	59.3	Other	19.5

Table 11: Comparison of annual marine vessel CO_2 emissions inventory estimates (billions of metric tonnes).

Year	GFW (AIS + S1)	GFW (AIS)	EDGAR	OECD	IMO
2017	1.11	0.824	0.881	-	1.06
2018	1.14	0.861	0.887	-	1.06
2019	1.17	0.884	0.875	0.887	-
2020	1.2	0.879	0.798	0.852	-
2021	1.27	0.941	0.833	0.891	-
2022	1.34	1	0.837	0.909	-
2023	1.38	1.05	0.89	0.917	-
2024	1.47	1.14	0.86	0.97	-
2025	1.53	1.19	-	-	-

6.2 Table S6

6.3 Table S7

6.4 Figure S7

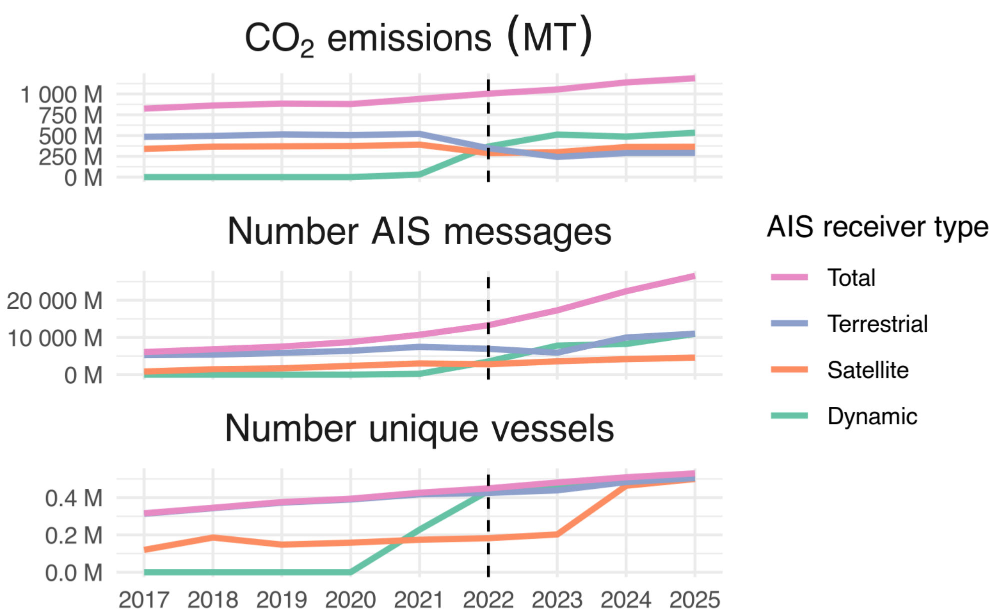


Figure 23: Annual total CO_2 emissions (metric tonnes) by AIS-broadcasting vessels, AIS pings (i.e., individual vessel timestamped location messages), and number of unique AIS-broadcasting vessels. Trends are disaggregated by AIS receiver type (satellite, terrestrial, and dynamic), and also shown as a total aggregated across receiver types. A dashed vertical line is shown for 2022, the first year in which dynamic AIS was widely adopted.

6.5 Figure S8

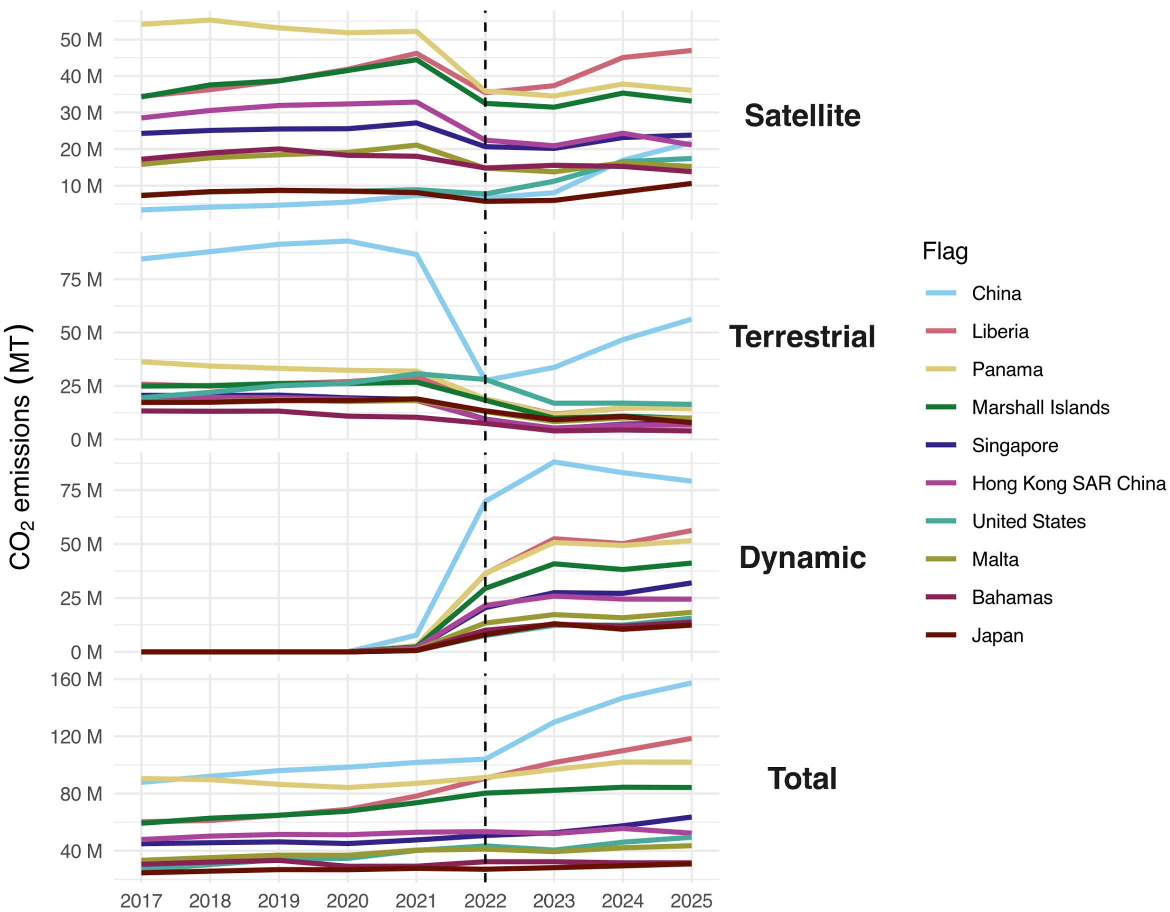


Figure 24: Annual total CO_2 emissions (metric tonnes) by AIS-broadcasting vessels, disaggregated by AIS receiver type (satellite, terrestrial, and dynamic) and total aggregated across receiver types, for the top 10 flags with the most emissions in 2025. A dashed vertical line is shown for 2022, the first year in which dynamic AIS was widely adopted.

6.6 Figure S9

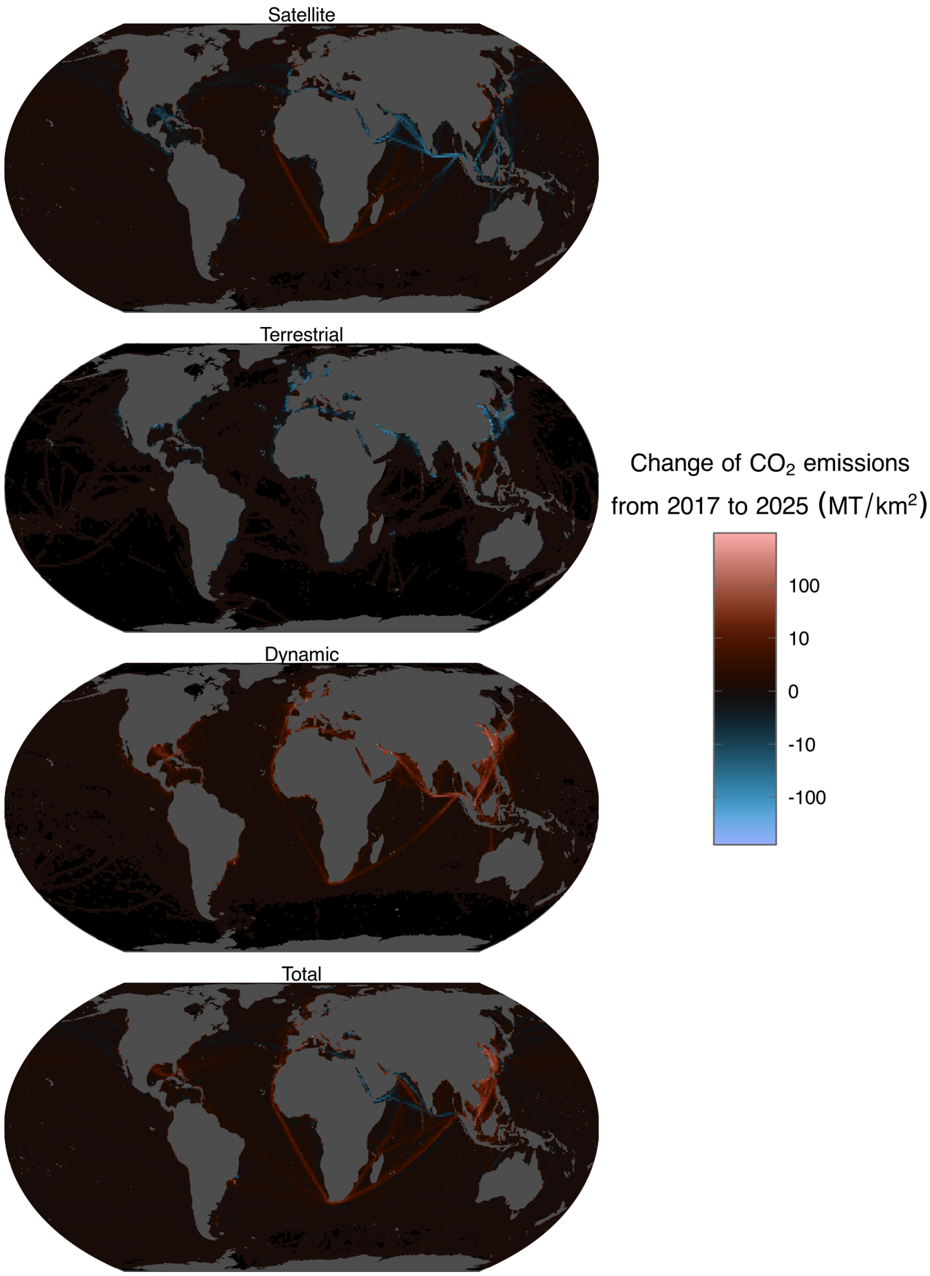


Figure 25: Spatial change in CO_2 emissions by AIS-broadcasting vessels between 2017 and 2025 (metric tonnes) by receiver type (satellite, terrestrial, and dynamic) and total across receiver types, at a 1x1 degree spatial resolution; emissions are area-normalized for each pixel (metric tonnes per square kilometer, shown on a pseudolog-10 scale).